

✓ Traditional competition is:

1/1

- ☐ both internal and external challenge for a company.
- ☐ not a challenge for a company.
- ☐ an internal challenge for a company.
- ☒ external challenge for a company. ✓

✓ Batch size reduction leads to:

1/1

- ☐ decreased production costs in the case of classical manufacturing systems.
- ☐ increased production costs in the case of CIM systems.
- ☒ increased production costs in the case of classical manufacturing systems. ✓

✓ Which of the following is/are not component(s) of a CIM system?

1/1

- ☐ CAD
- ☒ CAN ✓
- ☐ CAQ
- ☐ ERP

✓ Decreasing the distances between the machines of a production system usually:

1/1

- ☐ reduce the productivity
- ☐ increase the setup time
- ☒ increase the productivity ✓
- ☐ reduce the setup time

✓ The smaller the manufacturing-space ratio, the less efficient the manufacturing system is.

1/1

☒ true



☐ false

✗ CAM is used to control:

0/1

☐ the transport systems

☐ the robots in a factory

☐ the automated storage and retrieval systems

☒ the machine tools



Correct answer

☒ the machine tools

☒ the robots in a factory

☒ the transport systems

☒ the automated storage and retrieval systems

✗ Work-in-process inventory is:

0/1

☐ safety stock

☐ movement stock

☐ anticipation stock

☒ cycle stock



Correct answer

☒ movement stock

✗ Which of the following statements is/are not true?

0/1

- ☐ The quality standards force the manufacturers to produce to certain standards involving investments in quality assurance.
- ☒ The quality standards offer a lower credibility on the market and a potential decrease in sales. ✓
- ☐ The quality standards offer a higher credibility on the market and a potential decrease in sales.
- ☒ The quality standards offer a higher credibility on the market and a potential increase in sales. ✗

Correct answer

- ☒ The quality standards offer a lower credibility on the market and a potential decrease in sales.
- ☒ The quality standards offer a higher credibility on the market and a potential decrease in sales.

✓ Setup time is not the time required to get a machine ready for production. 1/1

- ☒ false ✓
- ☐ true

✓ Computer integrated manufacturing involves?

1/1

- ☐ lower flexibility
- ☐ larger stocks
- ☐ higher production costs
- ☒ better control of the production process ✓

✓ Computer Aided Process Planning (CAPP):

1/1

- ☐ increases planning time
- ☒ decreases human intervention ✓
- ☐ decreases flexibility

✓ The shortest manufacturing time is found in the production system of type:1/1

- ☐ repetitive
- ☒ continuous ✓
- ☐ line
- ☐ project
- ☐ job shop

✗ A four stations production system can have:

0/1

- ☐ no routing
- ☒ fixed routing ✓
- ☐ mixed routing
- ☒ variable routing ✓

Correct answer

- ☒ fixed routing
- ☒ variable routing
- ☒ no routing
- ☒ mixed routing

✓ What type of MS is used for producing chemical products?

1/1

- ☐ job shop
- ☐ line
- ☐ project
- ☐ repetitive
- ☒ continuous



✓ Assembling a product requires 10 minutes when it is done for the first time 1/1 by a trained worker. The process will be repeated to make 10 products. If the learning rate is 85%, what will be the delivery time of the 100th product? ($m=-0.23$, $10^m=0.59$).

- ☐ 2.85
- ☐ 59
- ☒ 5.9
- ☐ 0



✓ If the setup time of a manufacturing system increases:

1/1

- ☐ the productivity is not related to the setup time
- ☐ the productivity increases
- ☒ the productivity decreases



✗ The cycle stock exists when the sales are larger than the production.

0/1

- ☐ false
- ☒ true



Correct answer

- ☒ false

✓ The large batch production is not efficient in:

1/1

☒ modern manufacturing systems

✓

☐ classical manufacturing systems

✓ A project manufacturing system can be used if:

1/1

☐ the design drawings are simple

☒ the customer is identified before the production starts

✓

☐ the total time for production is short

☒ products are complex

✓

✗ Early detection of faults may increase significantly the production costs. 0/1

☒ true

✗

☐ false

Correct answer

☒ false

✓ What is the best manufacturing strategy for a mature product whose demand can be forecasted? 1/1

☒ make to stock ✓

☐ engineering to order

☐ make to order

☐ assembly to order

✓ In a CIM system: 1/1

☒ the quality of the delivered products increases ✓

☒ the size of the production batches decreases ✓

☐ production costs increase

☒ inventories decrease ✓

☒ production processes are better controlled ✓

✗ Make to stock strategy is used: 0/1

☐ if the manufacturing lead time is more than the customer lead time

☒ if the manufacturing lead time is less than the customer lead time ✗

☒ if the demand can be forecast accurately ✓

Correct answer

☒ if the demand can be forecast accurately

☒ if the manufacturing lead time is more than the customer lead time

✓ Utilization of a facility depends on:

1/1

☐ the level of integration

☒ actual quantity produced during a given period

✓

☐ mean time to repair

☒ production capacity for a given period

✓

✓ According to J.T. Black, the price of parts and materials is not 50% of the selling price. 1/1

☒ true

✓

☐ false

✓ Work-in-process (WIP) is the quantity of parts or products:

1/1

☒ being processed

✓

☐ finished

☐ delivered

☒ being between processing operations

✓

✓ Enterprise resource planning (ERP) is efficient in CIM.

1/1

☐ false

☒ true

✓

✓ What is the meaning of the ratio quantity produced / production capacity? 1/1

- ☐ mean time between failures
- ☐ it has no meaning
- ☐ availability
- ☒ utilization ✓
- ☐ average manufacturing lead time

✓ A factory produces mobile phones and tablet PCs. On the line that produces mobile phones 5 models with minor differences can be made. On the line that produces tablets PCs, 10 models with major differences are produced. What is the total amount of products made by the company, if 1000 phones of each model and 500 tablets PC of each model are produced? 1/1

- ☐ 12500
- ☐ 1500
- ☒ 10000 ✓

✓ An operation that joins two or more components to create a new entity is: 1/1

- ☐ a storage operation
- ☐ a transport operation
- ☒ an assembly operation ✓
- ☐ a processing operation

✓ Quality assessment is more effective if done:

1/1

☐ at the beginning of the production cycle.

☒ during the production cycle.



☐ at the end of the production cycle.

✓ Which of the following statements is / are not true?

1/1

☐ the costs associated with material handling must be as small as possible.

☒ the costs associated with material handling must be as large as possible.



☒ for moving the large and heavy components, autonomous guided vehicles are recommended.



✓ Global economy is an internal challenge for a company.

1/1

☐ true

☒ false



✓ Guidance based on painted strips is the most indicated technology.

1/1

☒ false



☐ true

✓ Which of the following factors are considered when designing a material handling system? 1/1

- ☒ routing ✓
- ☒ plant layout ✓
- ☒ the material characteristics ✓
- ☒ flow rate ✓
- ☒ scheduling ✓

✓ In a vehicle based transport system the delivery cycle time depends on: 1/1

- ☒ unloading time ✓
- ☒ loading time ✓
- ☒ travel time ✓

✓ The setup time can be ignored in the case of: 1/1

- ☒ mass production systems ✓
- ☐ job shop production systems
- ☐ project production system

✓ The cheapest technology for vehicle guidance is: 1/1

- ☒ paint stripes ✓
- ☐ self guided vehicles
- ☐ imbedded guide wires

✓ To reduce the inventory levels:

1/1

☒ the residence time must decrease ✓

☐ the residence time must increase

☒ the number of inventory turns must increase ✓

☐ the number of inventory turns must decrease

✗ Which of the following statements is / are true?

0/1

☐ A part producer assemble parts.

☐ A vertically integrated plant only produce the parts.

☒ A pure assembly plant produce no part. ✓

☒ A vertically integrated plant only assembles the products. ✗

Correct answer

☒ A pure assembly plant produce no part.

- ✓ Each side of a two-aisle ASRS contains 10 storage compartments in the length direction and 10 compartments vertically. All storage compartments will be the same size to accommodate standard size pallets of dimensions: $x = 1$ m, $y = 1$ m. The height of a unit load is $z = 1$ m. What is the storage capacity of the ASRS? 1/1

☒ 400 ✓

☐ 100

☐ 200

☐ 10

- ✓ Which of the following statements is / are true? 1/1

☐ Less total space is required in a storage system that uses dedicated storage.

☐ Higher throughput can be achieved when a random storage strategy is implemented.

☒ Less total space is required in a storage system that uses randomized storage. ✓

☒ Higher throughput can be achieved when a dedicated storage strategy is implemented. ✓

- ✓ The setup time can be ignored when determining the operation cycle time for job shop production. 1/1

☐ true

☒ false ✓

✗ Each aisle of a two-aisle ASRS contains 5 storage compartments in the length direction and 5 compartments vertically. All storage compartments will be the same size to accommodate standard size pallets of dimensions: $x = 1$ m, $y = 1$ m. The height of a unit load is $z = 1$ m and the allowances are $a = 0.1$ m, $b = 0.1$ m, and $c = 0.1$ m. What is the width of the ASRS? 0/1

☐ 3.3

☐ 9.9

☒ 13.2 ✗

☐ 6.6

Correct answer

☒ 6.6

✗ Which of the following statements is / are true? 0/1

☐ The availability of ASRS is not a measure of its reliability.

☒ In a storage system the accessibility increases as the density decreases. ✓

☐ In the case of an ASRS its robot arm moves to each basket

Correct answer

☒ In a storage system the accessibility increases as the density decreases.

☒ In the case of an ASRS its robot arm moves to each basket

✓ The increase of the setup time increases the productivity of a manufacturing system. 1/1

☐ true

☒ false ✓

✓ Random storage strategy is more appropriate when: 1/1

- ☐ items are stored according to activity level
- ☒ the throughput is not an essential factor to be considered ✓
- ☒ the storage capacity must be used at maximum ✓
- ☐ items are stored according to their activity-to-space ratio
- ☐ items are stored in part number sequence

✓ What changes can be made to increase or decrease production capacity over long term? 1/1

- ☐ change the number of hours worked per shift
- ☒ increase the production rate, by making improvements in methods or process technology ✓
- ☐ change the number of shifts per week
- ☒ increase the number of work centers in the shop ✓

✓ The handling of materials must be performed: 1/1

- ☒ with no damage to the materials ✓
- ☒ accurately ✓
- ☒ safely ✓
- ☒ in a timely manner ✓
- ☒ efficiently ✓
- ☐ at high cost

✗ What is the meaning of the ratio $\text{production capacity} / \text{quantity produced}$? 0/1

☒ availability

✗

☐ mean time between failures

☐ average manufacturing lead time

☐ utilization

☐ it has no meaning

Correct answer

☒ it has no meaning

✗ Usually in a storage system the accessibility increases as the density increases:

0/1

- ☐ accessibility and density are not interrelated
- ☐ false
- ☒ true

✗

Correct answer

- ☒ accessibility and density are not interrelated

✓ Which of the following associations is / are correct?

1/1

- ☐ polycode sequence $\Leftrightarrow$ monocode sequence
- ☒ chain-type sequence $\Leftrightarrow$ polycode sequence
- ☒ monocode sequence $\Leftrightarrow$ hierarchical sequence
- ☐ hierarchical sequence $\Leftrightarrow$ chain-type sequence

✓

✓

✗ Which of the following associations is / are correct?

0/1

- ☐ production equipment $\Leftrightarrow$ machine groups
- ☐ similar parts $\Leftrightarrow$ part cells
- ☒ production equipment $\Leftrightarrow$ machine cells
- ☒ similar parts $\Leftrightarrow$ part families

✓

✓

Correct answer

- ☒ similar parts $\Leftrightarrow$ part families
- ☒ production equipment $\Leftrightarrow$ machine groups
- ☒ production equipment $\Leftrightarrow$ machine cells

✓ Which encoding structure provides the highest information density? 1/1

- ☒ monocode ✓
- ☐ hybrid
- ☐ none of the above
- ☐ polycode

✓ The complexity of a product is given by: 1/1

- ☒ the number of components ✓
- ☐ the number of operations required for the sale

✗ Which of the following are specific to mass production systems? 0/1

- ☒ cheap products ✓
- ☒ simple products ✓
- ☐ complex products
- ☐ expensive products
- ☐ complex drawings
- ☐ simple drawings

Correct answer

- ☒ simple products
- ☒ simple drawings
- ☒ cheap products

✓ Which of the following statements is / are true when the work flow in the cell is analyzed: 1/1

☐ PIM should be as low as possible

☒ PBM should be as low as possible ✓

☒ PIM should be as high as possible ✓

☐ PBM should be as high as possible

✓ The use of group technology: 1/1

☒ reduces the work-in-process ✓

☐ increases the setup time.

☒ reduces the manufacturing lead time. ✓

☐ increases the number of operations in the production process.

☒ reduces the number of operations in the production process. ✓

✗ What is the number of levels required to represent the characteristics of a product, if a 4 digits polycode structure is considered? 0/1

☐ 5

☐ there is no multilevel representation

☐ 3

☒ 4 ✗

Correct answer

☒ there is no multilevel representation

✗ Dedicated storage is used when products are not stored according to the level of activity. 0/1

☐ false

☒ true

✗

Correct answer

☒ false

✓ A processing operation: 1/1

- ☐ joins two or more components to create a new entity.
- ☒ transforms a work material from one state of completion to a more advanced state that is closer to the final desired part or product. ✓

✗ The bottleneck station is: 0/1

- ☐ the station with the lowest utilization.
- ☒ the station with the highest workload per server ✓
- ☐ the station with the smallest workload per server
- ☐ the station with the highest utilization.

Correct answer

- ☒ the station with the highest utilization.
- ☒ the station with the highest workload per server

✓ A dedicated flexible manufacturing system: 1/1

- ☒ is designed to produce a limited variety of part styles ✓
- ☐ is more appropriate when the part family is large
- ☒ all the part styles to be made on the system is known in advance. ✓

✓ Which of the following are specific to project - type production systems? 1/1

- ☐ high volume
- ☒ the design drawings are complex ✓
- ☒ the lead time is long ✓
- ☐ noncomplex products

✓ Which of the following is used when material must be moved in relatively large quantities between specific locations over a fixed path? 1/1

- ☐ AGVs
- ☐ cranes and pulleys
- ☒ conveyor systems ✓

✓ In CIM, manufacturing batch size reduction leads to increased production costs. 1/1

- ☒ false ✓
- ☐ true

✗ Which of the following is / are related to the new part test? 0/1

- ☒ Can the system process different part styles in a non batch mode? ✗
- ☐ Can the system accept changes in production scheduling?
- ☐ Can the system recover from equipment malfunctions without complete disruption of production?
- ☒ Can new part designs be introduced into the existing product mix with relative ease? ✓

Correct answer

- ☒ Can new part designs be introduced into the existing product mix with relative ease?

✓ In a flexible manufacturing system: 1/1

- ☐ manufacturing lead time increases.
- ☒ fewer machines are required. ✓
- ☒ direct labor requirements are reduced. ✓
- ☐ the responsiveness to change is lower

✓ To be flexible a manufacturing system must:

1/1

☒ quick changeover the operating instructions



☐ quick changeover the physical price

☒ distinguish different part or product styles



✗ Dedicated storage is used when:

0/1

☐ items are stored in part number or product number sequence

☐ items are stored according to their activity-to-space ratios

☒ items are not stored according to activity level,



Correct answer

☒ items are stored in part number or product number sequence

☒ items are stored according to their activity-to-space ratios

FINAL

0 of 0 points

✓ The design of a material handling system does not depend on the type of manufacturing system where it is embedded. 1/1

☐ true

☒ false ✓

✗ Design for life cycle refers to the product during the manufacturing and includes factors ranging from product delivery to product disposal. 0/1

☒ true ✗

☐ false

Correct answer

☒ false

✓ What type(s) of design is/are related to the concurrent engineering? 1/1

☒ design for life cycle ✓

☒ design for cost ✓

☒ design for quality ✓

☐ design for selling

✓ The direct connection between CAD and CAM is: 1/1

☐ CAQ

☐ CAE

☒ CAPP ✓

✓ Bottleneck station is the station with:

1/1

- ☐ the average workload per server
- ☒ the highest workload per server
- ☐ the lowest workload per server

✓

✓ Which of the following are related to the use of CAPP?

1/1

- ☒ lead time decreases
- ☐ lead time increases
- ☒ productivity increases
- ☒ processes are standardized

✓

✓

✓

✓ What is the right order of the general steps required for the management activities of an investment project? 1/1

- A. the equipment is installed and debugged;
- B. vendor progress in building the equipment is monitored;
- C. proposal to justify the investment is prepared;
- D. management approvals are granted for the investment;
- E. vendor quotations are solicited;
- F. training of operators;
- G. responsibility for running the equipment is turned over to the operating department.
- H. order is placed to the winning vendor;
- I. any special tooling and supplies are ordered;

- ☐ E, B, D, G, C, F, A, H, I
- ☐ D, E, B, C, G, A, F, H, I
- ☐ A, B, C, D, E, F, G, H, I
- ☐ C, A, B, E, D, F, H, I, G

☒ C, D, E, H, B, I, A, F, G

✓

✓ Processes Planning is always made automatically in a company. 1/1

☒ false ✓

☐ true

✓ The maximum production rate of all parts is the maximum production rate of the bottleneck station. 1/1

☒ true ✓

☐ false

✓ An FMS needs less time than a classical manufacturing system to adapt to a new product. 1/1

☒ true ✓

☐ false

✓ Which of the following statements is / are correct? 1/1

- ☒ Order scheduling assigns the production orders to the various work centers in the plant. ✓
- ☒ Order progress module monitors the status of the various orders in the plant. ✓
- ☒ Order release provides the documentation needed to process a production order through the factory. ✓

✓ Which of the following statements is / are not correct regarding manufacturing resource planning (MRP II)? 1/1

- ☒ inventories increase ✓
- ☐ inventories decrease
- ☐ better collaboration with suppliers
- ☐ better collaboration with customers

✗ In production planning and control, what types of inventory are of greatest interest? 0/1

- ☐ finished products
- ☐ components sold
- ☒ work in process ✓
- ☐ raw materials

Correct answer

- ☒ raw materials
- ☒ work in process
- ☒ finished products

✗ What is the correct order of the three phases of a typical shop floor control system: 0/1

- A. order release
- B. order progress
- C. order scheduling

☐ A, C, B

☐ B, C, A

☐ A, B, C

☐ B, A, C

☐ C, B, A

☒ C, A, B ✗

Correct answer

☒ A, C, B

✓ Master production planning deals with the allocation of human resources. 1/1

☐ true

☒ false ✓

✓ Which of the following questions are considered in the decision to make or buy a product? 1/1

☒ What is the anticipated product life? ✓

☒ Is the process available in-house? ✓

☒ What is the total production quantity? ✓

☒ How do part costs compare? ✓

✓ What type(s) of card(s) is/are used in Kanban systems?

1/1

- ☐ for the authorization of the planning
- ☐ for the authorization of the quality control
- ☒ for the authorization of the delivery ✓
- ☐ only one type used for both authorization and delivery of the production
- ☒ for the authorization of the production ✓

✗ For the polycode structure, the number of digits of the code is determined by the number of levels in the hierarchy. 0/1

☐ false

☒ true ✗

Correct answer

☒ false

✓ If a manufacturing system must process different part styles in a non batch mode what flexibility test does it pass? 1/1

☐ schedule change

☐ new part

☐ error recovery

☒ part variety ✓

✓ The Just-in-Time production system operates successfully if: 1/1

☒ the setup time of the production line is small ✓

☐ the setup time of the production line is large

☐ the production is made in large size batches

☒ the production is made in small size batches ✓

☐ equipment with high reliability are not used

✓ In the case of chase production strategy:

1/1

☐ the number of hours worked is constant.

☒ the number of employees is variable. ✓

☐ the number of employees is constant.

☒ the number of hours worked is variable. ✓

✓ House of quality transforms the customer requirements in characteristics of the manufactured products 1/1

☒ true ✓

☐ false

✓ In classical production systems, manufacturing batch size reduction leads to increased production costs. 1/1

☒ true ✓

☐ false

✓ Modern Quality Control focuses on inspection of the products:

1/1

☒ during the production process ✓

☐ only at the beginning of the production process

☐ after they are produced

✓ In the case of "Just-in-Time" manufacturing systems, inventories represented by WIP are zero:

1/1

☐ true

☒ false



✓ Which of the following inspection techniques is most suitable for a production line that produces toothbrushes?

1/1

☒ acceptance sampling by lots



☐ 100% inspection

✓ Quality can not be defined as the degree of excellence of a product.

1/1

☒ false



☐ true

✗ The quality of a product can be assessed based on its:

0/1

☒ performance



☐ features

☒ durability



☒ serviceability



☒ aesthetic appeal



Correct answer

☒ performance

☒ features

☒ serviceability

☒ aesthetic appeal

☒ durability

✓ What tests should pass a manufacturing system to be considered flexible? 1/1

- ☒ part variety test ✓
- ☒ error recovery test ✓
- ☒ new part test ✓
- ☐ costs test
- ☐ productivity test

✗ Traditional quality control (QC): 0/1

- ☐ is performed fully automated
- ☒ is performed only before the product is manufactured ✗
- ☐ involves the use of statistical quality control techniques
- ☐ is performed for each step of the production process
- ☒ is the responsibility of the inspection department ✓

Correct answer

- ☒ is the responsibility of the inspection department
- ☒ involves the use of statistical quality control techniques